

Replication & Analysis of the **DEPM**

Investigating discrepancies, validating methodology, and extending the architecture for robust energy forecasting.

INITIAL PHASE RECAP

- 🚀 Implemented DEPM (ResNet + DNN + XGBoost)
- 📊 UCI Household Power Consumption (2M+ rows)
- ⚠️ Discrepancies found with reported metrics

2M+

DATA SAMPLES (47 MONTHS)

98%

PAPER CLAIMED ACCURACY

$P=V \times I$

PHYSICS LEAKAGE RISK

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DISCOVERY

Critical Flaw: Data Leakage

The paper's methodology includes a feature that is mathematically equivalent to the target.

⚠️ The Model is Cheating

It learns arithmetic, not energy patterns.

THE PHYSICS RELATIONSHIP

$$P = V \times I$$

Variable P

Target: GAP



Variable I

Feature: GI



Result

Leakage!

Logistic Regression

SIMPLE

99.4%

Accuracy with Leakage

DEPM (Paper)

COMPLEX

98.0%

Reported Accuracy

Preprocessing & Feature Pipeline

Robust feature engineering implemented to provide context beyond raw sensor readings without introducing leakage.

 Dimensionality Reduction

95% Variance

10 Components

StandardScaler + PCA applied to reduce noise and feature space.

 Sub-Metering Aggregation

$$\text{SubTotal} = \text{Sub1} + \text{Sub2} + \text{Sub3}$$

Total sub-metering usage

 Proportional Ratios

$$\text{SubRatio}_i = \text{Sub}_i / (\text{SubTotal}+1)$$

Usage distribution by zone

 Temporal Features

$$\begin{aligned} \text{Hour} &= (0-23) \\ \text{DayOfWeek} &= (0-6) \\ \text{Season} &= (\text{M}\%12//3)+1 \end{aligned}$$

Cyclical time patterns

 Contextual Flags

$$\begin{aligned} \text{IsWeekend} &= 1 \text{ (Day} \geq 5) \\ \text{IsHoliday} &= 1 \text{ (Public)} \end{aligned}$$

Behavioral differentiation

Paper Credibility Analysis

⚠ Statistically Impossible Pattern

Precision	= Accuracy - 0.01
Recall	= Accuracy + 0.01
F1-Score	= Accuracy

This identical pattern repeats in 40+ rows across all 6 tables.

1

The 54-Sample Matrix

Claims 2M+ dataset, but Figure 14 sums to **N=54**. Fabricated evaluation.



Plagiarized Text

Mentions "diagnosis values" (Medical) & "image histogram" (CV). Paper-mill signs.



No Reproducibility

"Code availability: Not applicable". Impossible to verify claims.

REPLICATION RESULTS

Full Replication: With vs Without Leakage

● WITH LEAKAGE (PAPER)

Accuracy Range

95-99%

Features Used

~40+ incl. Global_intensity

<> depm_final.ipynb

● WITHOUT LEAKAGE (OURS)

Accuracy Range

85-93%

Features Used

~15 Clean Features

<> depm_final_noleak.ipynb

VS

Replicated Scope: All 6 Experiments

T1:
Activation

T2:
Hyperparams

T3:
Models

T4:
Optimizers

T5: XGBoost
LR

T6: Batch
Size

DEPM Variants: The Hybrid Question

10 Models Trained

 STANDALONE DL	Base Architecture	 DEPM HYBRID	+ ResNet + XGBoost
DNN Deep Neural Network BASE		DEPM-DNN Paper's Proposed HYBRID	
LSTM Long Short-Term Memory BASE		DEPM-LSTM Extension HYBRID	
BiLSTM Bidirectional LSTM BASE		DEPM-BiLSTM Extension HYBRID	
GRU Gated Recurrent Unit BASE		DEPM-GRU Extension HYBRID	
BiGRU Bidirectional GRU BASE		DEPM-BiGRU Extension HYBRID	

Does the DEPM hybrid (adding XGBoost) actually improve performance?



Cross-Dataset Validation



RESIDENTIAL

Household Power

SOURCE
UCI

SAMPLES
2M+ Rows

DOMAIN
Residential

TARGET
GAP Split



INDUSTRIAL

Steel Industry

SOURCE
UCI #851

SAMPLES
35,040

DOMAIN
Industrial

TARGET
Load Type



Key Finding: Generalization Confirmed

The DEPM pipeline and architecture perform robustly across different energy domains and data scales.

DEPLOYMENT

Production Dashboard

FastAPI Prediction Application



Model Selector (10 variants)



Custom Input Mode



Quick Hour Slider



24h Daily Profile

FastAPI

PyTorch

XGBoost

Chart.js



http://localhost:8000/predict

PREDICTION VALUE

0.847 kW

Model: DEPM-BiLSTM
Time: 14:00

MODEL ACCURACY



24H LOAD PROFILE



METRICS

Acc: 0.92

Prec: 0.93

Rec: 0.91

F1: 0.92

STRATEGIC PIVOT

From Classification to Continuous Forecasting

WHY PIVOT?

Binary "High/Low" is useless for grid operators. They need real values.

NEW FOCUS: REGRESSION

Predicting actual **kW values** using state-of-the-art time-series forecasting.

Proposed Architecture: CNN-BiLSTM



Input: 24h Sequence

Hourly resampled data (Industry Standard STLF)



1D CNN Module

Extracts local spatial-temporal features from consumption patterns



Bi-LSTM Module

Captures forward & backward long-term dependencies



Output: Continuous kW

Inverse-transformed predictions for real-world usage

● MAE: **0.32** kW (326 W)

● RMSE: **0.47** kW

● MAPE: **41.3**%

SUMMARY

The Complete Project Journey

01



Critical Analysis

Identified data leakage & impossible metrics

02



Rigorous Replication

With & without leakage + cross-dataset validation

03



Architecture Extension

Built 10 DEPM variants + FastAPI dashboard

04



Strategic Pivot

Binary class. → Continuous regression

05



New Implementation

Production-ready CNN-BiLSTM model



KEY TAKEAWAY

A proper energy management system requires continuous load forecasting, not binary classification. We transitioned from exposing a flawed approach to implementing a scientifically valid model.